

Zhen Zhang

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Education

University of California, Santa Barbara (UCSB)

Sept. 2023 – Present

Ph.D. in Computer Science, advised by Prof. [Xin Eric Wang](#)

California, USA

- **Research Interests:** Large Language Models, Reinforcement Learning, Efficient Learning, and Multimodality.
- **Awards:** UCSB Computer Science Outstanding Publication of the Year Award, 2026.

Tsinghua University

Sept. 2019 – June 2023

B.Sc. in Physics

Beijing, China

- **Awards:** Academic Excellence Scholarship, 2021 and 2022.

Publications

* indicates equal contribution.

Zhen Zhang, Changyi Yang, Zijie Xia, Zhen Yang, Chengzhi Liu, Zhaotiao Weng, Yepeng Liu, Haobo Chen, Jin Pan, Chenyang Zhao, Yuheng Bu, Alkesh Patel, Zhe Gan, Xin Eric Wang. “[Length Value Model: Pretraining Value Model for Scalable Length Prediction and Control.](#)” COLM 2026.

Zhen Zhang, Kaiqiang Song, Xun Wang, Yebowen Hu, Weixiang Yan, Chenyang Zhao, Henry Peng Zou, Haoyun Deng, Sathish Reddy Indurthi, Shujian Liu, Simin Ma, Xiaoyang Wang, Xin Eric Wang, Song Wang. “[CM2: Reinforcement Learning via Checklist Reward for Multi-Turn Multi-Step Agentic Tool Use.](#)” arXiv.

Zhen Zhang*, Xuehai He*, Weixiang Yan, Ao Shen, Chenyang Zhao, Shuohang Wang, Yelong Shen, Xin Eric Wang. “[Soft Thinking: Unlocking the Reasoning Potential of LLMs in Continuous Concept Space.](#)” NeurIPS 2025.

Zhen Zhang, Yifan Yang, Kai Zhen, Nathan Susanj, Athanasios Mouchtaris, Siegfried Kunzmann, Zheng Zhang. “[MaZO: Masked Zeroth-Order Optimization for Multi-Task Fine-Tuning of Large Language Models.](#)” EMNLP 2025.

Zhen Zhang, Jialu Wang, Xin Eric Wang. “[Parameter-Efficient Cross-lingual Transfer Learning of Vision and Language Models via Translation-based Alignment.](#)” EMNLP 2023.

Shengding Hu*, **Zhen Zhang***, Ning Ding, Yadao Wang, Yasheng Wang, Zhiyuan Liu, Maosong Sun. “[Sparse Structure Search for Parameter-Efficient Tuning.](#)” NeurIPS 2022.

Jiabao Ji*, Bairu Hou*, **Zhen Zhang***, Guanhua Zhang*, Wenqi Fan, Qing Li, Yang Zhang, Gaowen Liu, Sijia Liu, Shiyu Chang. “[Advancing the Robustness of Large Language Models through Self-Denoised Smoothing.](#)” NAACL 2024 Short.

Yifan Yang, **Zhen Zhang**, Rupak Vignesh Swaminathan, Jing Liu, Nathan Susanj, Zheng Zhang. “[SharpZO: Hybrid Sharpness-Aware Vision Language Model Prompt Tuning via Forward-Only Passes.](#)” NeurIPS 2025.

Yujia Qin, Shengding Hu, ..., **Zhen Zhang**, ..., Tongshuang Wu, Heng Ji, Zhiyuan Liu, Maosong Sun. “[Tool Learning with Foundation Models.](#)” ACM Computing Surveys.

Shengding Hu, Ning Ding, Weilin Zhao, Xingtai Lv, **Zhen Zhang**, Zhiyuan Liu, Maosong Sun. “[OpenDelta: A Plug-and-play Library for Parameter-efficient Adaptation of Pre-trained Models.](#)” ACL 2023.

Wanjing Huang, Weixiang Yan, **Zhen Zhang**, Ambuj Singh. “[APEX: Empowering LLMs with Physics-Based Task Planning for Real-time Insight.](#)” arXiv.

Chengzhi Liu, Yuzhe Yang, Kaiwen Zhou, Zhen Zhang, Yue Fan, Yanan Xie, Peng Qi, Xin Eric Wang. “[Presenting a Paper is an Art: Self-Improvement Aesthetic Agents for Academic Presentations.](#)” ICLR 2026.

Jing Xiong, Qi Han, Yunta Hsieh, Hui Shen, Huajian Xin, Chaofan Tao, Chenyang Zhao, Hengyuan Zhang, Taiqiang Wu, **Zhen Zhang**, Haochen Wang, Zhongwei Wan, Lingpeng Kong, Ngai Wong. “[MMFormalizer: Multimodal Autoformalization in the Wild.](#)” [arXiv](#).

Zhaotian Weng, Antonis Antoniadis, Deepak Nathani, **Zhen Zhang**, Xiao Pu, Xin Eric Wang. “[Group-Evolving Agents: Open-Ended Self-Improvement via Experience Sharing.](#)” [arXiv](#).

Ruiyao Liu, Hui Shen, Ping Zhang, Yunta Hsieh, Yifan Zhang, Jing Xu, Sicheng Chen, Junchen Li, Jiawei Lu, Jianing Ma, Jiaqi Mo, Qi Han, **Zhen Zhang**, Zhongwei Wan, Jing Xiong, Xin Wang, Ziyuan Liu, Hangrui Cao, Ngai Wong. “[MathGen: Revealing the Illusion of Mathematical Competence through Text-to-Image Generation.](#)” [arXiv](#).

Internship Experience

ByteDance Seed, Post-training

June 2026 – Now

Student Researcher, advised by [Hongyi Guo](#) and [Lin Yan](#)

- Working on Value Pretraining and RL Scaling.

Zoom, GenAI

June 2025 – Sept. 2025

Research Intern, advised by [Kaiqiang Song](#) and [Song Wang](#)

- Worked on post-training methods for tool-use agents, focusing on reinforcement learning with checklist-based rewards for multi-turn, multi-step agentic tool use.

Microsoft STCA, WebXT-Search&Distribution

Apr. 2024 – Sept. 2024

Machine Learning Engineer Intern, advised by [Gong Ming](#) and [Wenbiao Ding](#)

- Optimized Bing Search models for TL;DR generation in RAG scenarios. Built pipelines for high-quality training data generation and applied DPO to improve model reliability when search results were insufficient or unanswerable.